

Sequential Circuits: Latches and Flip-Flop

6.1.1 Introduction to Sequential Circuits

In digital electronics, a combinational circuit is one of the essential types of circuits which is made up of basic gates. In the combinational circuit, the output at any instant of time depends only on the inputs at that time. Such circuits are called circuits with no memory because the output does not depend on the previous state of the circuit.

One more very important logical circuit is known as a sequential circuit, in which the present output depends not only on the present inputs but also on the previous output. When the output is generated, that output is stored in a storage unit known as a memory element. This stored output is fed back to the combinational circuit along with other inputs. The storage element, that is, memory, is capable of storing binary information in the form of 0 and 1.

There are two types of storage elements: latches and flip-flops. Both these storage elements are bi-stable devices. Bi means two. So, they have two stable states and can store either logic 0 or 1. Hence, these two storage elements are also known as 1-bit storage elements. The major difference between a latch and a flip-flop is how they are triggered. The latch or flip-flop's state can be changed via a momentary change in its input signal, and this momentary change is called a trigger. This trigger is provided by means of a clock pulse. When a trigger pulse is applied to the input, the output changes, and the storage device is said to be triggered. There are two ways of triggering: level triggering and edge triggering.

There are two types of level triggering:

1. High-level sensitive triggering
2. Low-level sensitive triggering

There are two ways to edge triggering:

1. Positive edge triggering
2. Negative edge triggering

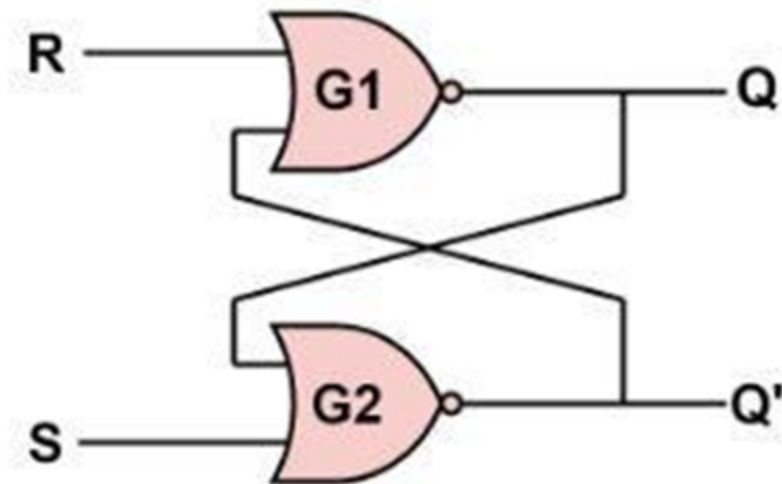
Latches are level sensitive, that is, the output changes only during the high voltage period or low voltage period. On the other hand, flip-flops are edge-triggered. In positive edge triggering, the output changes at the positive edge of the clock, whereas in negative edge triggering, the output changes at the negative edge of the clock.

Sequential circuits are used in many applications. It is used in the implementation of registers.

Registers are used in microprocessors and microcontrollers to store temporary data. It is used in the implementation of memory. Memory elements are typically implemented using flip-flops. Sequential circuits are also used in the implementation of analog-to-digital and digital-to-analog converters, counters, etc.

6.1.2 Storage Elements: RS Latch

The latch is a basic storage element that is capable of storing 0 or 1 by either setting or resetting. RS latch is also known as the reset-set latch. This latch has two inputs, R and S, and two outputs, Q and Q'. R is reset, S is set. Reset means storing 0 into the latch and set means storing 1 into the latch. This latch can be implemented either using NOR gates or NAND gates. In NOR gate implementation of the RS latch, it uses two 2-input NOR gates, G1 and G2. NOR gate truth table is shown in Figure 1.



NOR Gate Truth Table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Figure 1: NOR gate implementation

R input is connected to one of the inputs of G1, and the other input to G1 comes from the Q' output. Similarly, the S input is connected to one of the inputs of G2, and the other input comes from the Q output.

To demonstrate the functioning of the RS latch, consider the following four cases.

Case 1: $R = 1$ and $S = 0$

When one or both inputs of NOR gate is 1, the output will be 0. Hence, the output Q will be at 0. This Q output is fed as an input to G2. Since both the inputs to G2 are 0, the output will equal 1. Hence, for $R = 1$, and $S = 0$, $Q = 0$ and $Q' = 1$. Latch is said to be reset to 0.

Case II: $R = 0$ and $S = 1$

Since one of the inputs to G2 is 1, the output will equal 0. Hence, $Q' = 0$. Now G1 has both its inputs 0, so the output will be equal to 1. So, for $R = 0$, and $S = 1$, Q is equal to 1, and Q' is equal to 0. Hence, the latch is set to 1.

Case III: $R = 0$ and $S = 0$

Assume $Q = 0$. Since $Q = 0$, Q' will be equal to 1. Since the second input of G1 is 1, output $Q = 0$, and since both the inputs of G2 are zero, the output Q' will be 1. Hence, the output is the same as its previous state. For the same third case, let us assume Q is equal to 1 and Q' is equal to 0. Since both the inputs of G1 are 0, output Q is 1, and since one of the inputs of G2 is one, the output Q' will be 0.

Case IV: $R = 1$ and $S = 1$

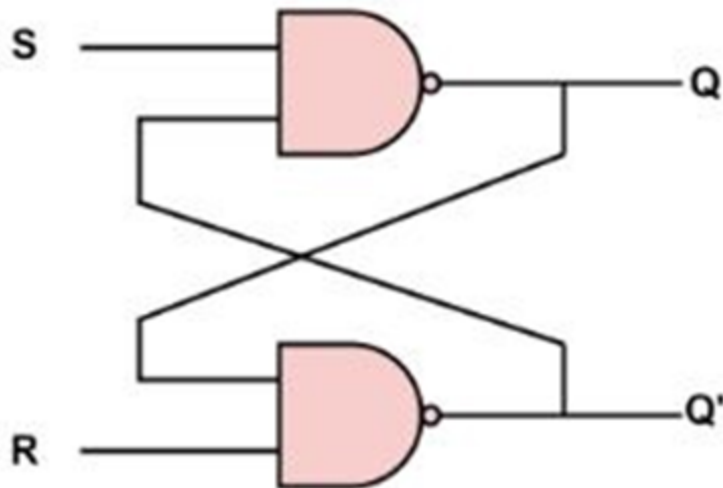
Both the NOR gate have one of their inputs at logic 1. So, both the NOR gates yield logic 0, that is, Q equals 0 and Q' equals 0. Having both Q and Q' at the same level is not correct. Hence, setting $R = 1$ and $S = 1$ is forbidden. The NOR gate RS latch truth table is shown in Figure 2.

R	S	Q
0	0	Previous state (Q_0)
0	1	1 ((Set)
1	0	0 (Reset)
1	1	Invalid ($Q = Q' = 0$)

Figure 2: Truth table for NOR gate RS Latch implementation.

When $R = 0$ and $S = 0$, the latch is said to be in the previous state. This state of latch is known as the memory state. When $R = 0$ and $S = 1$, latch is set to 1. When $R = 1$ and $S = 0$, latch is reset to 0. When $R = 1$ and $S = 1$, latch will be in a forbidden state or invalid state where both Q and Q' will be holding the same state.

Figure 3 shows the NAND gate implementation of the RS latch and the corresponding truth table.



NAND Gate RS Latch Truth Table

R	S	Q
0	0	Invalid ($Q = Q' = 0$)
0	1	0
1	0	1
1	1	Previous state (Q_0)

Figure 3: NAND gate RS Latch implementation

When $R = 0$ and $S = 0$, latch will be in an invalid state. When $R = 0$ and $S = 1$, the latch is set to 0. When $R = 1$ and $S = 0$, the latch is reset to 1. When $R = 1$ and $S = 1$, the latch is said to be in the previous state.

6.1.3 Storage Elements: RS Flip-Flops

The RS flip-flop is one of the most basic sequential logic circuits. The RS stands for SET RESET. Similar to RS latch, the RS flip-flop is a one-bit memory and is a bi-stable device. Flip-flop word means that it can be “FLIPPED” into one logic state or “FLOPPED” back into another. RS flip flop has three inputs. The first input is called “RESET,” which will reset the flip-flop to 0 and is labeled as R. The second input is known as “SET,” which will set the flip-flop to 1 and is labeled as S. The third input is a clock. Clocking causes the flip-flop either to change or to retain its output signal based on the values of the input signals at the transition. The main difference between RS Latch and the RS flip-flop is how the device is triggered. RS latch is level triggered, whereas RS flip-flop is edge triggered. It can be triggered at the positive edge of the clock or at the negative edge of the clock. Figure 4 shows the clocked RS flip-flop, which consists of a basic NOR RS latch and two AND gates.

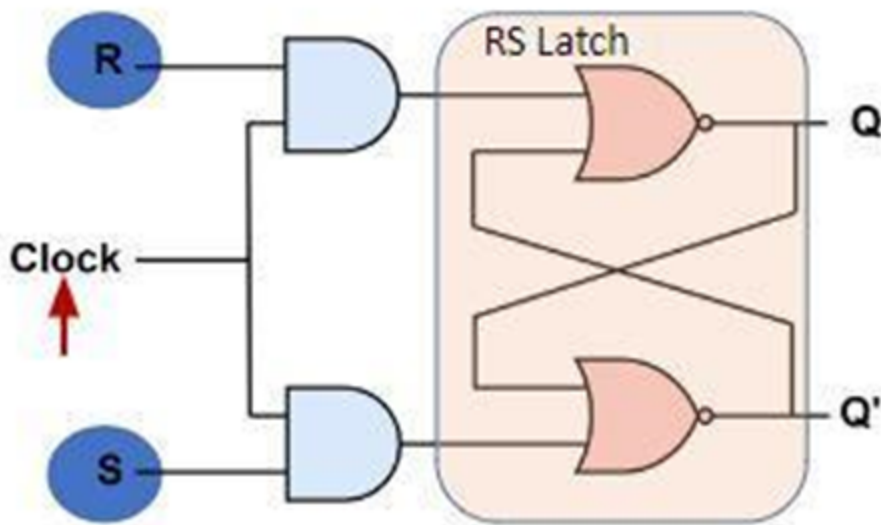


Figure 4: Clocked RS flip-flop

R and S inputs are applied to one of the inputs of these two AND gates, and another input of these gates is connected to the clock. The outputs of the two AND gates remain at 0 if the clock pulse is zero, irrespective of R and S input values.

When the clock input is 1, the input R and S values are applied to the basic RS latch. The flip-flop is set to 1 when $R = 0$, $S = 1$, and $\text{Clock} = 1$. The flip-flop is reset to 0 when $R = 1$, $S = 0$, and $\text{Clock} = 1$. When $R = 0$, $S = 0$, and $\text{Clock} = 1$, the flip-flop is said to be in the previous state. In this state, the outputs of the flip-flop, Q and Q' will be at the previous logic level. When $S = 1$, $R = 1$, and $\text{Clock} = 1$, the flip-flop is indeterminate, or the flip-flop is said to be in the forbidden or invalid state. The truth table of the RS flip-flop is shown in Figure 5.

R	S	Q
0	0	Previous state (Q_0)
0	1	1 ((Set)
1	0	0 (Reset)
1	1	Invalid ($Q = Q' = 0$)

Figure 5: Clocked RS flip-flop

6.1.4 Storage Element: JK Flip-Flops

The R-S flip-flop has the disadvantage of meta-stability; that is, if both inputs go high, the resulting state is unpredictable. To overcome this problem, a JK flip-flop is used. JK flip-flop is the most widely used of all the flip-flop designs and is a universal flip-flop circuit. This flip-flop was developed by Jack

Kilby. The role of inputs J and K are the same as that of R and S inputs. Logic 1 on J input sets the flip-flop, and logic 1 on K input resets or clears the flip-flop. A clocked J K flip-flop is as shown in Figure 6. This flip-flop uses a basic SR flip-flop and additional AND gates. Q' output is ANDed with J and clock inputs. To set a flip-flop, inputs J, Q' should be 1, along with an active clock pulse. Similarly, Q input is ANDed with K and Clock inputs so that the flip-flop is cleared during a clock pulse only if Q was previously 1. The structure and truth tables are given in Figures 6 and 7, respectively.

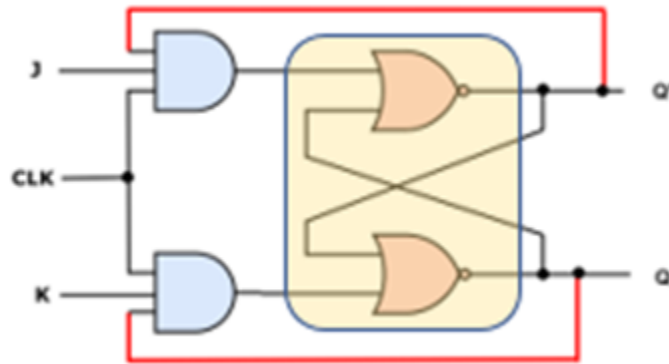


Figure 6: Structure of JK flip-flop

Data Inputs		Outputs		Inputs to SR Flip-Flop		Output Q_{n+1}
J	K	Q_n	Q_n'	S	R	
0	0	0	1	0	0	0
		1	0	0	0	1
0	1	0	1	0	0	0
		1	0	0	1	0
1	0	0	1	1	0	1
		1	0	0	0	1
1	1	0	1	1	0	1
		1	0	0	1	0

Figure 7: Truth table of JK flip-flop

J and K are the two data inputs to the JK flip-flop. Outputs Q_n and Q_n' represent the present output. S and R are the two inputs to the RS flip-flop. Q_{n+1} represents the next output. Let us assume that the clock is high when inputs to the flip-flop are to be applied.

Consider four cases of inputs J and K.

Case I: $J = 0$ and $K = 0$.

There are two possibilities for present outputs Q_n and Q_n' . Irrespective of these two possible values for present outputs, since both J and K are zero, the output of both the AND gates will be 0. So, S and R will be zero. Therefore, Q_{n+1} will be in the previous state. That is, when Q_n is equal to 0, Q_{n+1} will be zero, and when Q_n is 1, Q_{n+1} will be 1; hence, when $J = 0$, $K = 0$, the flip-flop will be in the previous state.

Case II: $J = 0$ and $K = 1$

Assume Q_n is 0 and Q_n' is 1. Since J is 0, the output of AND gate $G1$ is zero. Similarly, since Q_n is 0, AND gate $G2$ output is zero. Therefore, S and R are applied with 0, which makes the flip-flop remain in the previous state. Since the previous state of the flip-flop is 0, the next state Q_{n+1} is 0.

Now assume Q_n is 1 and Q_n' is 0. Since Q_n' is 0, the output of AND gate $G1$ is zero. So, S will be 0. Since all the inputs to $G2$ are 1, the output will be 1. Hence, R will be equal to 1. For $S = 0$ and $R = 1$, the output of the flip-flop will be reset to 0. So, Q_{n+1} is equal to 0. Therefore, for $J = 0$ and $K = 1$, the flip-flop is said to be cleared or reset.

Case III: $J = 1$ and $K = 0$

Assuming Q_n equals 0 and Q_n' equals 1, S and R become 1 and 0, respectively. This, in turn, sets the flip-flop to 1, which is 1. Assuming Q_n equals 1 and Q_n' equals 0, S and R become 0. When both S and R inputs are 0, a flip-flop will be in the previous state. In this case, the previous state of the flip-flop is 1. Hence, Q_{n+1} is set to 1.

Case IV: $J = 1$ and $K = 1$

Assuming Q_n equals 0 and Q_n' equals 1, S and R become 1 and 0, respectively. This, in turn, sets the flip-flop to 1, that is, Q_{n+1} is 1. Assuming Q_n equals 1 and Q_n' equals 0, S and R become 0 and 1, respectively.

When $S = 0$ and $R = 1$, the flip-flop will be reset to 0. As you can see, when $J = 1$, $K = 1$ flip-flop changes its state from 1 to 0 and 0 to 1. And the flip-flop is said to be in a toggle state.

6.1.5 Demerits of JK Flip-Flops

The invalid state of the RS flip-flop is eliminated by the JK flip-flop by using a feedback connection from output to input. In the JK flip-flop, when $J = 1$ and $K = 1$, the flip-flop is said to be in a toggle state. JK flip-flop also possesses a problem. This problem is known as a race-around condition. Consider the last row of the truth table given in Figure 7, where both J and K are equal to one. The output Q_{n+1} of the flip-flop depends on the status of previous outputs Q_n and Q_n' . Q_{n+1} will be 1 when Q_n is 0, and Q_{n+1} will be 0 when Q_n is 1. Let both J and K be equal to 1, and Q_n is zero, and clock pulse. Let t_p be the pulse width from the leading edge of the clock to the falling edge. Δt be the propagation time of a flip-flop. Propagation delay, in general, is defined as the length of time starting from when the input to a device is applied to the time that the output of the device is available. Since the previous output Q_n is zero, the present output toggles to 1. At the end of Δt , flip-flop finds the clock is high, the previous output is 1, and toggles to 0. This toggling back and forth between 0 and 1 continues during the entire clock period t_p . At the end of the clock pulse, the value of Q_{n+1} is uncertain. This condition is referred to as the "race-around condition." The solution to solve the problem is to keep t_p less than Δt . This solution is difficult to implement practically because ICs usually have very small propagation delays. A more feasible solution is to use a modified version of the JK flip-flop known as a master-slave flip-flop.

6.1.6 Storage Element: JK Master-Slave Flip-Flops

JK master-slave flip-flop is made up of two JK flip-flops. The first flip-flop is known as a master flip-flop which is positive edge triggered, whereas the second one is known as a slave which is negative edge triggered. These two flip-flops are called master and slave because the master responds to its J and K inputs before the slave flip-flop. Figure 8 shows the master-slave JK flip-flop.

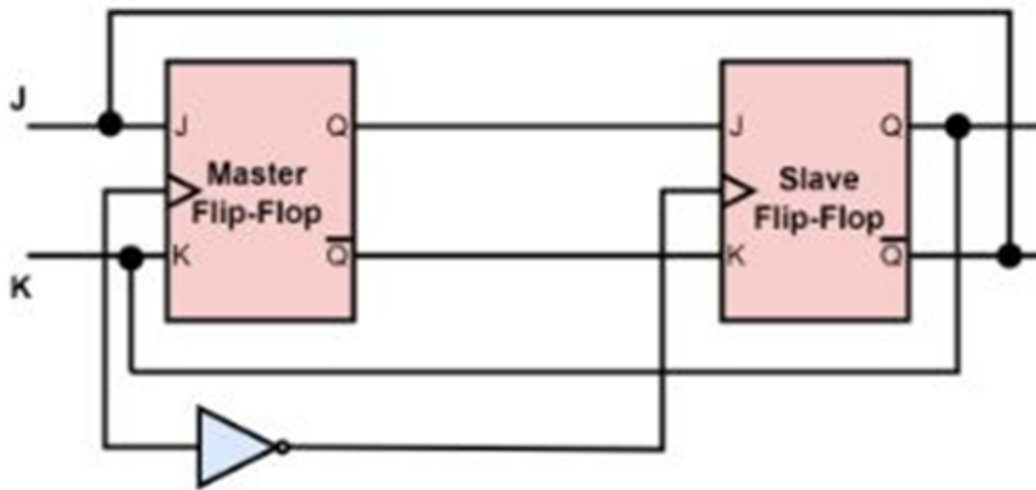


Figure 8: Truth table of master-slave JK flip-flop

Figure 9 is the truth table of the master-slave JK flip-flop.

J	K	Flop-Flop Output Q_{n+1}
0	0	Previous State
0	1	Reset
1	0	Set
1	1	Toggle

Figure 9: Truth table of Master-Slave JK flip-flop.

When both the inputs of the master are zero, the flip-flop is disabled, and the output of the master and slave remains unchanged. The flip-flop will be said to be in the previous state. If $J = 0$ and $K = 1$, the master resets on the positive edge of the clock. That is, Q output of the master will be 0. During the negative edge of the clock, the slave resets to 0. The slave copies the master again. The flip-flop is said to be in the reset state. If $J = 1$ and $K = 0$, the output of the master is set to 1 on the positive clock transition. The high Q output of the master drives the J input of the slave, so on the negative clock transition, the slave is set to 1. So, the slave copies the action of the master. When both J and K inputs of the master are high, the output toggles on the positive edge of the clock, and the slave sets or resets on the negative edge of the clock based on the master's output. When the master toggles during the positive edge of the clock and at the end of the clock, let us assume its output is 0. Now for the slave, J input of 0, and K input is 1, which makes the flip-flop reset on the negative edge of the clock. If the master output is 1 after toggling, the slave output will be set. So, slave will have stable output. This is how race condition is eliminated in JK master-slave flip-flop.

6.1.7 Storage Element: D Flip-Flops

The main drawback of RS flip is the "invalid state"; that is, the output becomes forbidden when S and R inputs are set to 1. Similarly, in the case of the JK flip-flop, setting $J = 1$ and $K = 1$ leads to a race

condition. The main purpose of these two flip-flops is to set or reset the flip-flop. In the case of the RS flip-flop, when $S = 1$ and $R = 0$, the flip-flop will be set to 1, and when $S = 0$ and $R = 1$, the flip-flop will be reset to 0. For these two cases, S is the complement of R . Similarly, in the case of the JK flip-flop, when $J = 1$ and $K = 0$, the flip-flop will be set to 1, and when $J = 0$ and $K = 1$, the flip-flop will be reset to 0. Similar to the RS flip-flop, here also, J is the complement of K . Therefore, instead of having two inputs, the flip-flop can have only one input, and the other input can be generated by complementing it.

D flip-flop is a combinational electronic circuit that has only two inputs, D and clock. The D input is also known as data input. It is a clocked flip-flop. D flip-flop implementation using RS and JK flip-flop is shown in Figure 10.

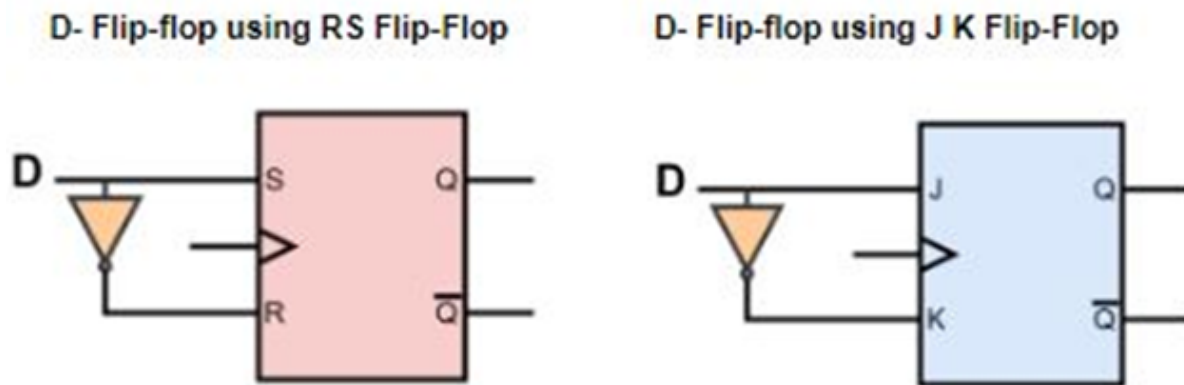


Figure 10: D flip-flop block diagram

In the case of D flip-flop implementation using Rs flip-flop, the D input is connected to the S input and its complement to R input. Similarly, in the case of implementation using $J K$ flip-flop, the D input is connected to J input and complement of D to K input. When the clock input is low, the output retains the original value stored in the flip-flop. When the D flip-flop is clocked, output Q follows the input D . The state of its output signal Q is delayed until the next rising edge of a clock occurs. Hence, this flip-flop is also known as a delay flip-flop. A truth table for the two implementations is shown in Figure 11.

Truth Table

Clock	D	Q_{n+1}
0	X	Q_n
1	1	1
1	0	0

Figure 11: D flip-flop block diagram

The D flip-flop symbol is shown in Figure 12.

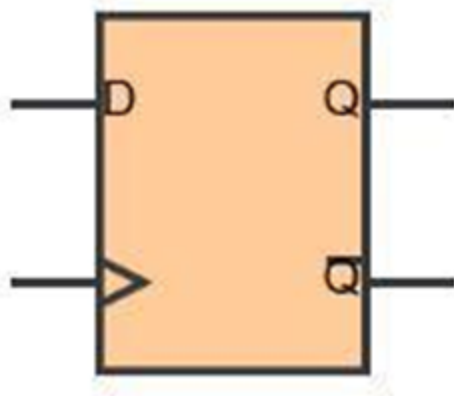
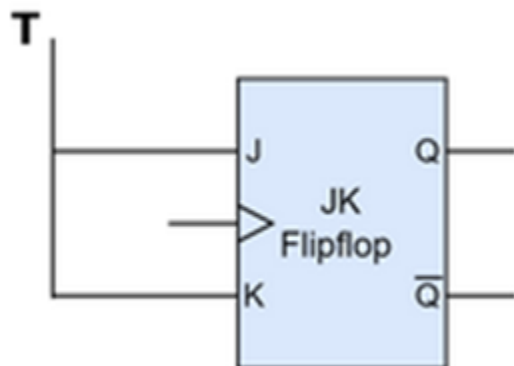


Figure 12: D flip-flop symbol

The main applications include the implementation of time delay circuits and counters. They are also used in shift registers for data transfer applications.

6.1.8 Storage Elements: T Flip-Flops

In JK flip-flop, when both J and K are equal to 1 flip-flop, it will be in a toggle state. That is, output changes from 0 to 1 and 1 to 0. In counter applications, J and K inputs are always at logic 1. This can be achieved by modifying the J K flip-flop. In T flip-flop, "T" denotes "Toggle." This flip-flop works as a toggle switch. That is, the next state of the flip-flop will be the complement of the present state. T Flip-flop has two inputs, "T" and clock. Two outputs, Q and Q'. For constructing a T flip-flop from JK flip, we need to tie inputs J and K together. Due to this T flip-flop is also known as single input JK flip-flop. There are two variants of T flip-flops based on the triggering applied at the clock input. In a positive edge-triggered flip-flop, output toggles at the positive edge of the clock pulse. In a negative edge-triggered flip-flop, output changes at the negative edge of the clock pulse. Figure 13 shows the logic circuit and truth table of the T flip-flop.



Truth Table:

Data Inputs	Past Outputs		Present Outputs	
T	Q_0	Q_0'	Q_{n+1}	Q'_{n+1}
0	0	1	0	1
0	1	0	1	0
1	0	1	1	0
1	1	0	0	1

Figure 13: T flip-flop and its truth table
T can have a value of either 0 or 1. When T = 0, the flip-flop will be in the previous state. When T = 1, output toggles from 0 to 1 and 1 to 0. One of the popular applications of T flip-flops is in the implementation of frequency divider circuits.

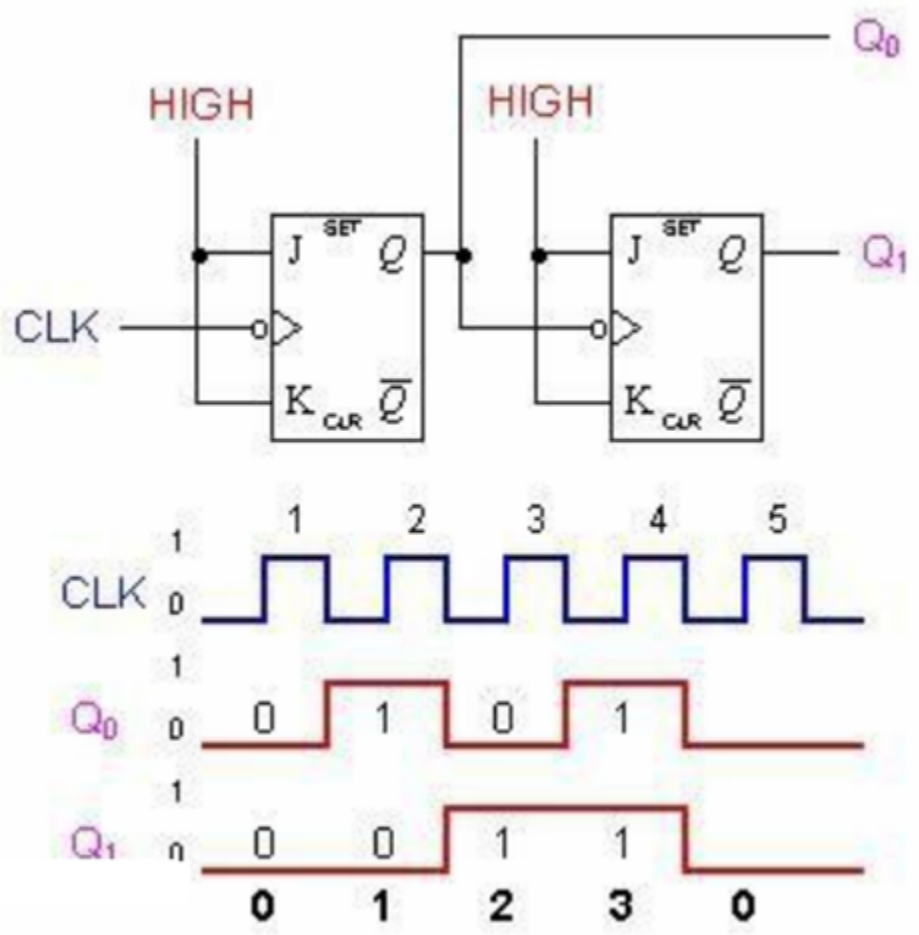


Figure 14: Frequency divider circuit

Figure 14 shows the frequency divider circuit, which uses two JK flip-flops. The inputs of both the flip-flops are supplied with logic High input. Both the flip-flops are negative edge triggered. So, the output changes during the negative edge of the clock. The clock input to the second JK flip-flop comes from the Q output of the first flip-flop. The output Q0 changes its output at the negative edge of the input clock pulse. Whereas the output of the second flip-flop changes its output at the negative edge of the Q0 output. So, if clock frequency is f , then output Q0's frequency is $0.5f$ and Q1's frequency is $0.25f$.

Registers

6.2.1 Introduction to Registers

Flip-flop is a sequential circuit that can store 1 bit of digital information. This flip-flop is considered as an essential element of a sequential logic system. Any sequential circuit can be designed using flip-flops and combinational circuits. The register contains an array of flip-flops that can store a word of information. The number of flip-flops in a register depends on the number of bits in a word.

Registers find their application in various digital systems, including microprocessors and microcontrollers. For example, the LC3 processor has a set of registers used to store temporary data required during the processing of an instruction. These sets of registers are known as register set or register file. Data can be entered into the register in serial form or in parallel form. In serial form, the data is entered one bit at a time, whereas in parallel form, all bits of the data are entered simultaneously. Data in serial form is also referred to as a temporal code, and in parallel form is called spatial code. For example, consider binary data D3 D2 D1 D0 equal to 0 1 0 1, where D3 is MSB and D0 is LSB. This data in serial form and parallel form is shown in Figure 15.

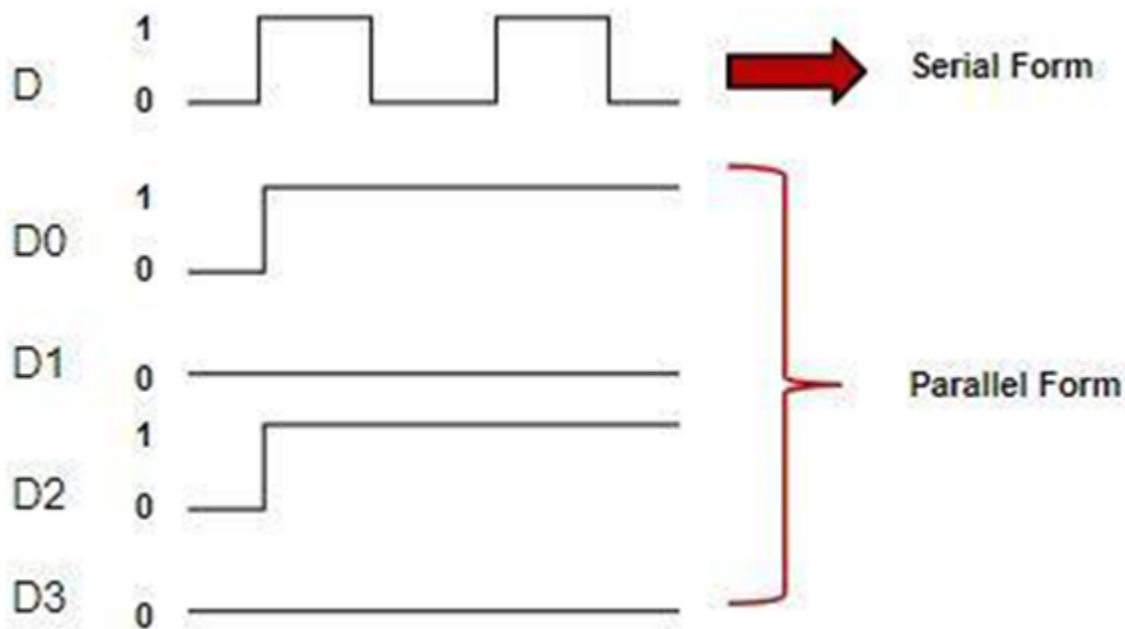


Figure 15: Data in serial and parallel forms

One data line is required for serial data transfer, whereas for parallel data transfer, multiple lines are required. Based on how data is entered and retrieved, registers are classified into four categories.

1. Serial in, Serial Out (SISO)
2. Serial in, Parallel Out (SIPO)
3. Parallel in, Serial Out (PISO)
4. Parallel in, Parallel Out (PIPO)

The register in which data are entered and taken out in serial or parallel form is known as a shift register. It is possible to design a register whose contents can be shifted toward the right or left.

Such registers are called right-shift registers and left-shift registers. In a bi-directional shift register,

data can be shifted from left to right as well as in reverse direction. A register that can implement all four types of register that is SISO, SIPO, PISO, and PIPO, is known as a universal register.

6.2.2 Types of Registers

Shift registers are one of the most commonly used registers in which data is entered and taken out in a serial or parallel form.

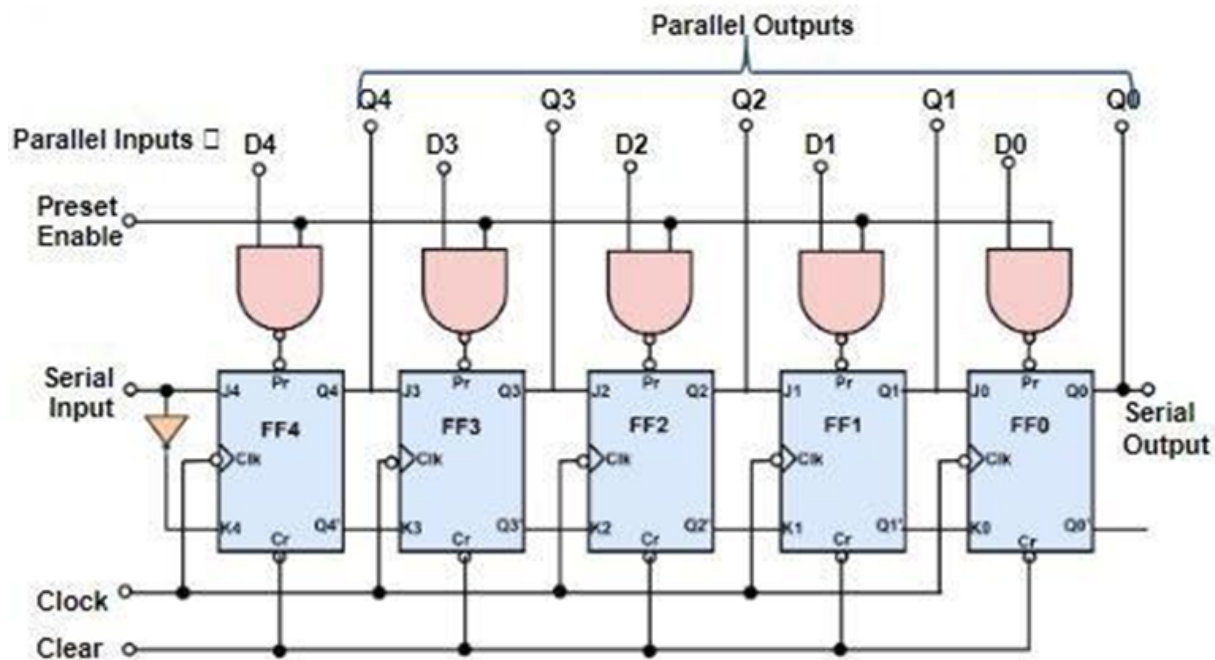


Figure 16: Block diagram of 5-bit shift register

Figure 16 shows the 5-bit register implemented using five master slave JK flip-flops. In such a register, flip-flops are cascaded where the output of one flip-flop is connected to the input of the next.

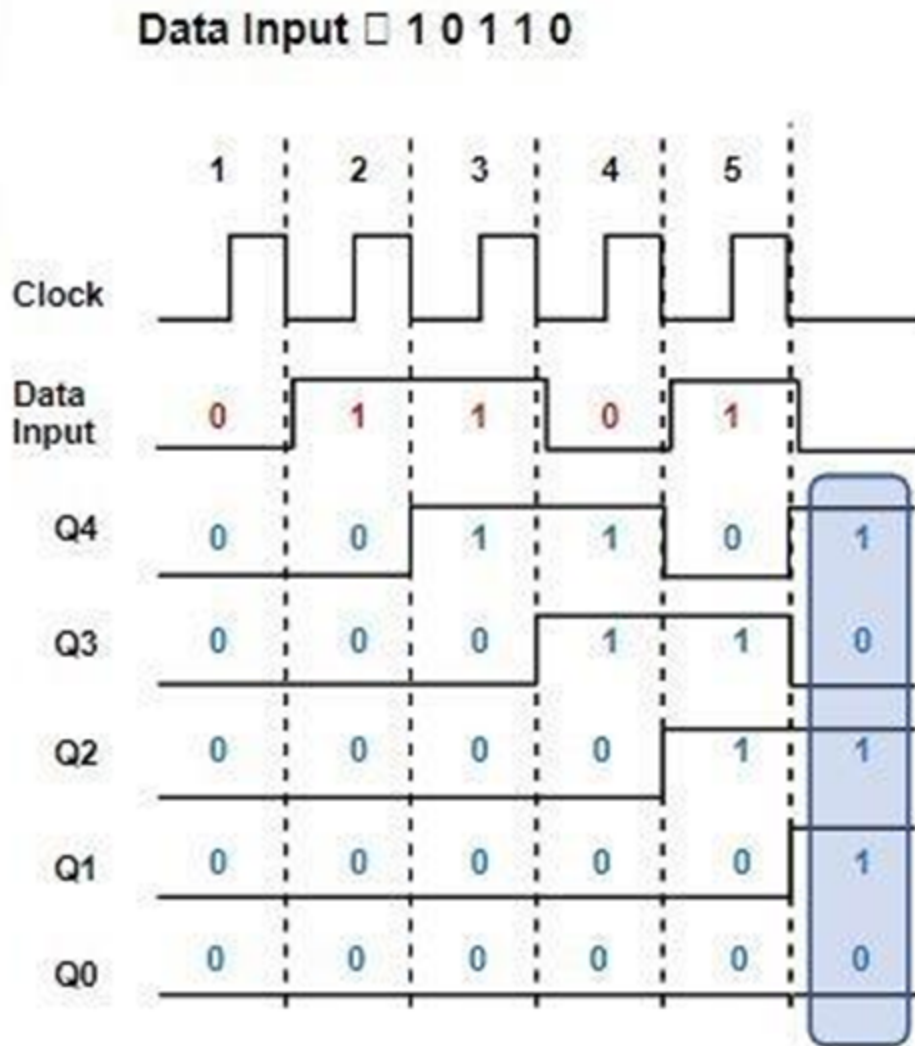


Figure 17: Waveform of the 5-bit shift register

To shift a 0 into the flip-flop, J is set to 0, and K is set to 1. To shift a 1 into the flip-flop, J is set to one, and K is set to zero. Initially, flip-flops are set to 0 by using Clear input. All the flip-flops are tied to a common clock, and flip-flops are activated at the negative edge of the clock pulse. Clock pulses are numbered T1, T2, T3, etc. Data to be shifted is loaded into the serial input line. The process of entering the data input starts with the data input corresponding to the least significant bit. Since initially, all the flip-flops are cleared, the outputs of Q0 to Q4 will be zero during T1. At the falling edge of the T1 clock pulse, the output of flip-flop 4 will be 0 since the data input is 0, and the outputs of all other flip-flops are 0 since their inputs are 0. At the falling edge of T2, input corresponding to the next bit one is applied, the flip-flop outputs will be Q4 equals 1, Q3, Q2, Q1, and Q0 equals 0. At the falling edge of T3, the next input 1 is applied. The flip-flop outputs will be Q4, Q3 equals 1, Q2, Q1, Q0 equals 0. At the falling edge of T4, the next input zero is applied. The flip-flop outputs will be Q4 equals 0, Q3, Q2 equals 1, Q1, Q0 equals 0. At the falling edge of T5, the next input 1 is applied. The flip-flop outputs will be Q4 equals 1, Q3 equals 0, Q2, Q1 equals 1, and Q0 equals 0. At the end of the fifth clock pulse, the outputs of flip-flops are Q4 equals 1, Q3 equals 0, Q2 equals 1, Q1

equals 1, and Q0 equals 0, which is the same as the number which was entered. Observe that the number of clock pulses required for entering the data is the same as the number of data bits. Data is entered into the register serially. Data can also be entered in parallel by using a parallel input line and preset enable.

Note: The process of entering the data is referred to as writing into the register. The process of retrieving the data is referred to as reading.

6.2.3 Applications of Registers

In general, registers are used for storing temporary data during the execution of instructions in a processor. A register may hold an instruction, the address of a storage unit, or any kind of data, such as a bit sequence or characters. Figure 18 shows the block diagram of a general-purpose processor.

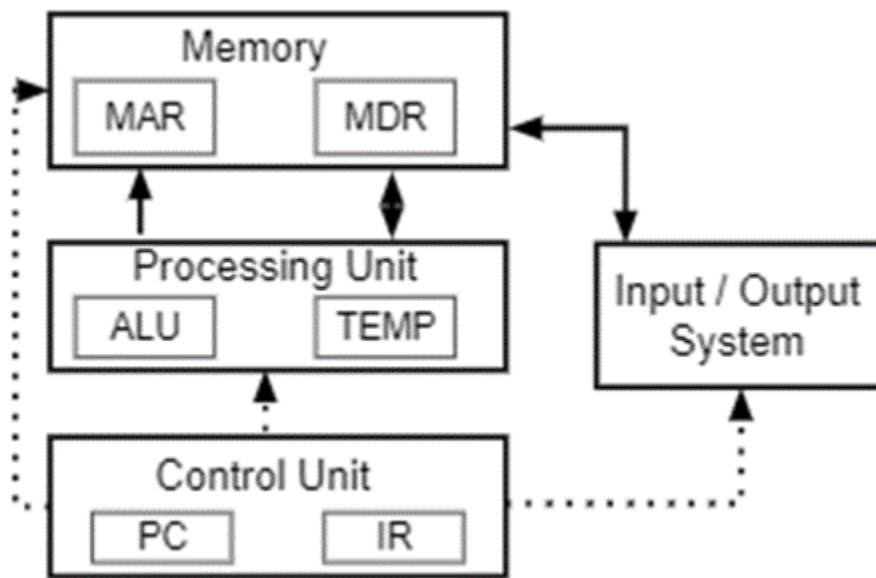


Figure 18: Block diagram of a processor

All processors have a register known as a program counter, also known as an instruction pointer. A program counter is used to store the address of the instruction that is to be executed next. The Instruction register is used to store the instruction fetched from memory. MDR (memory data register) and MAR (memory address register) act as a buffer between the memory and the processor. MDR holds the data, whereas MAR holds the address. Apart from these registers, there are registers that will aid during arithmetic and logical operations and address computation. These sets of registers are known as register files.

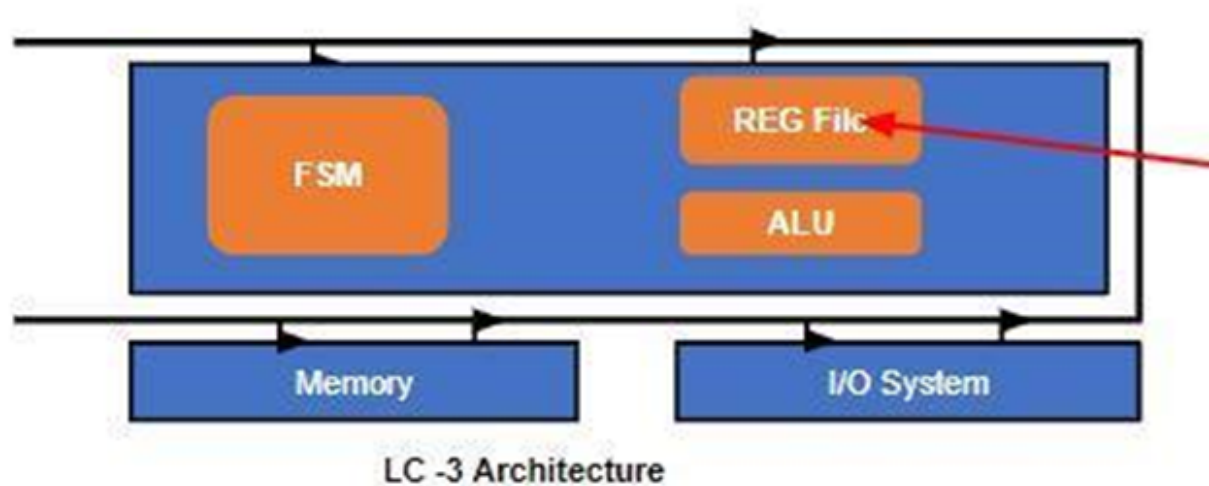


Figure 19: LC-3 Processor architecture

In the LC-3 computer of Figure 19, there is a register file containing several registers. Another application of the register is storing the status of the processor. The status information stored in the register is known as the processor status word (PSW). The purpose of using register storage is to improve the execution speed of a program since accessing the register is faster than the main memory.

While switching from one function to another, it is required to store the context of the calling function. Processors are usually connected to external devices either for data acquisition or data transfer. In such cases, it is essential to know the status of the device connected. For example, how does a computer know the ink cartridge is empty, or printer is not on, or there is no paper? This information is captured as a status in a register. Bits in the status register are examined for further action.

Another application of the register is to introduce a time delay in digital signals. A Serial Input, Serial Output register, may be used to introduce a delay. In data communication, parallel data transmission is used for shorter distances, providing greater speed. The best example is communication between the computer and a printer. On the other hand, serial data communication is suitable for transmitting data over longer distances. Computers generate parallel data that needs to be converted to serial data before it is transmitted. Similarly, at the receiving end, received serial data needs to be converted back to parallel data. For this purpose, parallel to serial converter and serial to parallel converter is needed. Parallel data can be converted to serial form using parallel in serial out shift register. In contrast, serial data can be converted to parallel data using serial in, parallel out shift register.

A special type of sequential circuit used to count the pulse is known as a counter. They are used to count specific events happening in the circuit. These counters are implemented using registers. Registers are also used to generate a prescribed sequence of bits. Such sequence generators find their application in counters, random bit generators, and code generators.

Memory

6.3.1 Concept of Memory: Address Space and Addressability

A flip-flop is one-bit memory capable of storing logical 1 and 0. The memory unit stores information that usually consists of program instructions which are coded in binary form, data to be processed and intermediate and final results. The basic element of memory is a flip-flop. Memories are usually made up of a large number of locations where data word is stored. Each location is identified by a unique identifier known as a memory address. The number of bits of information stored in each location is known as addressability. The total number of uniquely identifiable locations is known as address space. For example, 2 KB memory. K stands for a kilo. K stands for 1024.

Most memories are byte-addressable, where each location is capable of storing 8 bits of information. The modern computer, which is to perform large scientific calculations, has higher addressability. For example, 64-bit addressable memory.

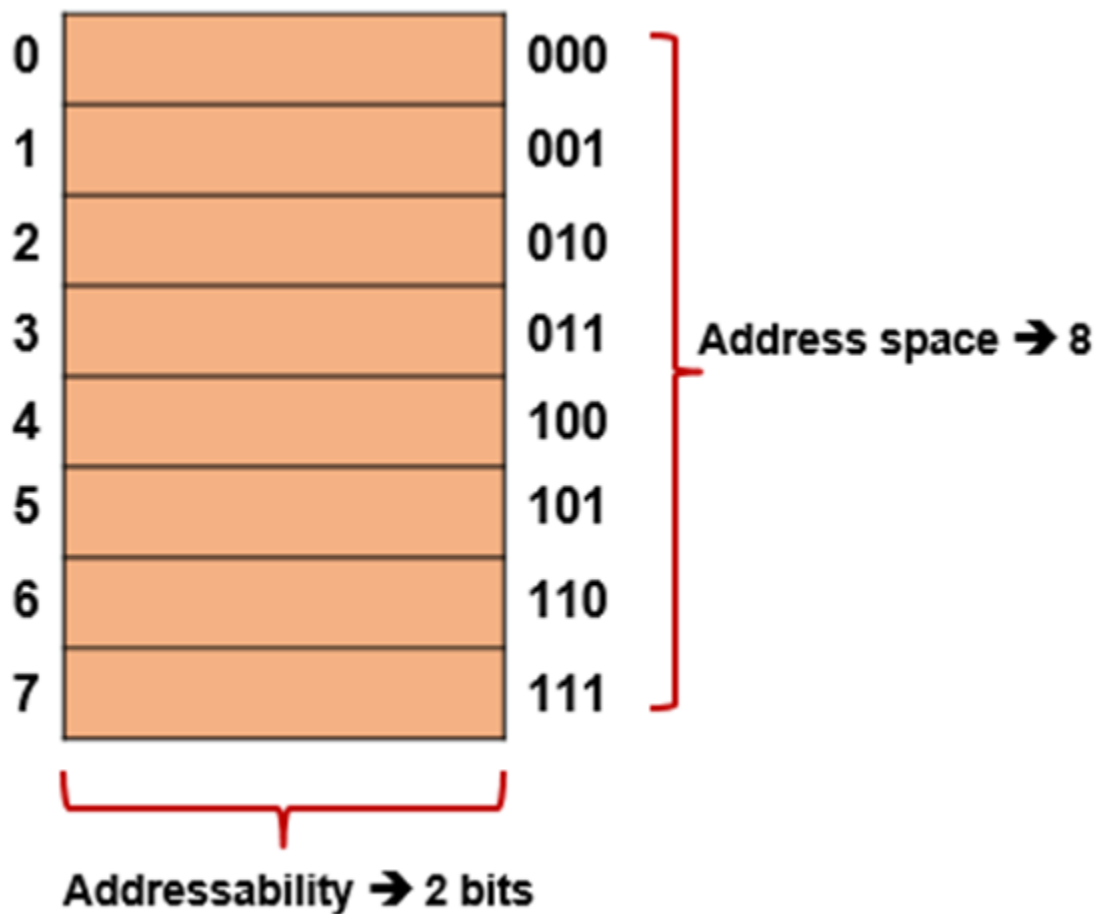


Figure 20: Block diagram of 8 × 2-bit memory

The simple memory organization of 8 × 2-bit memory is shown in Figure 20. The address space of this memory is 8, which means there are eight memory locations in the memory. These memory locations are uniquely addressed. The starting address of the memory is 0, and the last memory location is 7. The binary equivalent of these addresses is, as shown. Addressability is 2 bits; that is,

in each memory location, 2-bit information can be stored. There is a relationship between the number of memory locations and a number of bits needed to represent an address.

$$2^N = M$$

2

N

$$= M$$

where N is the number of address bits, and M is the total number of memory locations.

6.3.2 Types of Memory: Random-Access, Sequential-Access, Volatile, and Non-Volatile

Memory is an integral part of a computer that is required to store data, programs, and results. The memory of a computer is categorized into two types: primary memory and secondary memory.

Primary memory is the computer's main memory, where programs and data are stored temporarily.

Program and data stored in this main memory are directly accessed by the CPU during program execution. On the other hand, secondary memory is used to store large volumes of data

permanently on a secondary storage device. The secondary storage memory is permanent memory.

Depending on the type of material used, there are two most common types of memory

semiconductor memory and magnetic memory. Semiconductor memory is a faster, lightweight, compact, but expensive memory, whereas magnetic memory is slower and cheaper.

Information stored in memory can be accessed either sequentially or randomly. In sequential access memory, information is read or written sequentially. Access time is the amount of time it takes to read or write information. In the case of random-access memory, access time is independent of where information is stored.

Semiconductor memories are categorized into two types: RAM—random access memory and

ROM—read only memory. Originally RAM was known as read-write memory. RAM is volatile

memory, that is, contents will be erased in case of no power. Therefore, it is used as temporary

storage. RAM is usually packaged as a chip and may require two more chips to make a memory of

the required size. For example, there is a RAM chip called SN74LS207, which is 1024×1 -bit

memory. To make it 1024×8 bits, we need eight such chips. RAM comes in two categories: dynamic

RAM, in short, known as D-RAM, and static RAM, in short, known as SRAM.

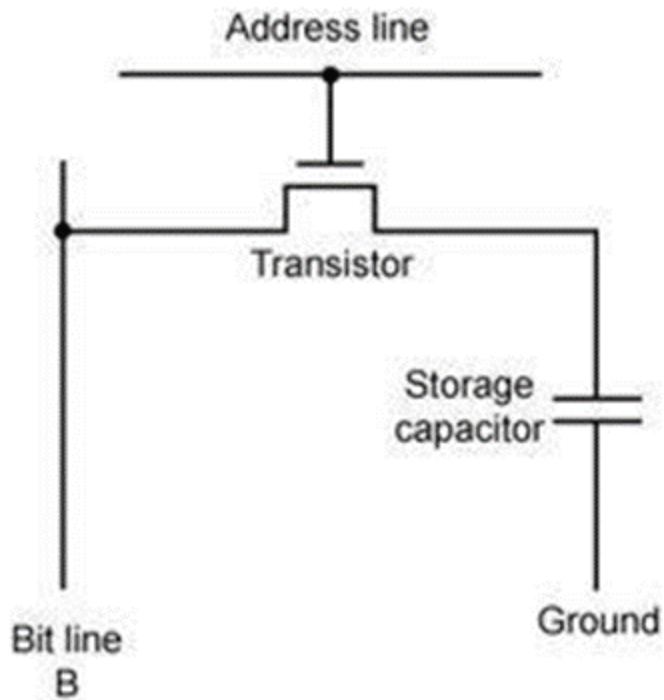


Figure 21: DRAM structure

Figure 21 shows the structure of D-RAM. It has a very simple construction. Each memory cell consists of a transistor and a capacitor. Here, the data is stored in the form of a charge on a capacitor. A charge cannot be held for a longer time. Charges tend to leak. Hence, there is a need for a refreshing circuit even when the circuit is powered. This is an extra overhead. DRAMS are smaller in size and less expensive but slower. DRAMS are extensively used in the implementation of main memory and frame buffers.

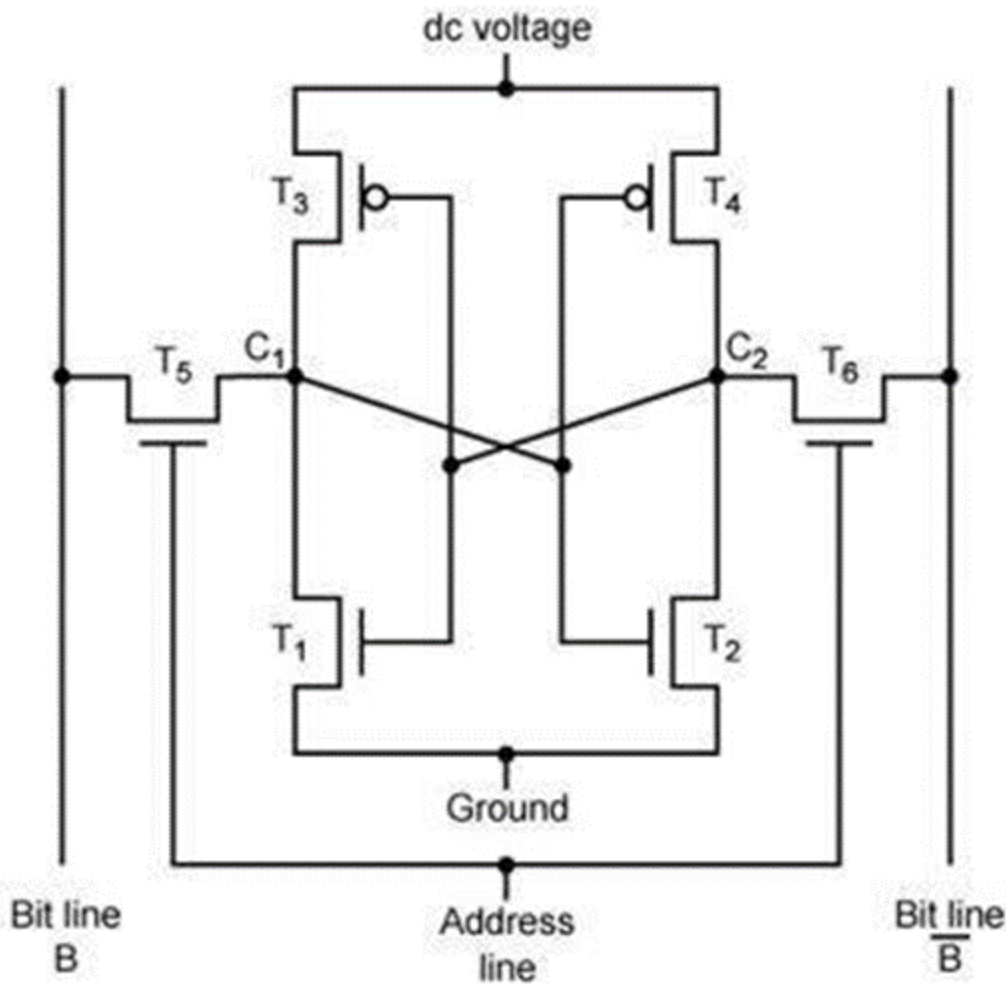


Figure 22: SRAM structure

The structure of SRAM is shown in Figure 22. Since there are no capacitors involved, no charges to leak. Hence, no refreshing circuit is needed. The structure of SRAM is complex compared to DRAM. Hence, it requires a larger space per bit. Also, it is more expensive than DRAM. Since it is faster compared to DRAM, it is used in the implementation of cache memory. Cache memory is a high-speed memory that acts as a bridge between the main memory and the CPU. There is a comparison table that compares SRAM and DRAM with reference to transistor per bit, access time, the requirement of refreshing circuit, cost, and applications.

Read only memory is also random-access memory. This permanent and non-volatile memory has a few variants. In ROM, memory contents are written during production, and there is no option for the user to program it. Note that the process of programming a ROM is sometimes called burning the ROM. Programmable ROM, PROM is a second variant where the user can program data only once. The difference between a PROM and a ROM is that a PROM is manufactured as blank memory, whereas a ROM is programmed during the manufacturing process. To write data onto a PROM chip, a special device called a PROM programmer or PROM burner is needed. The third variant is erasable PROM, EPROM. It is a special type of PROM that can be erased by exposing it to

ultraviolet light or X-ray. Once it is erased, it can be reprogrammed. The fourth variant is electrically erasable PROM, EEPROM. This has an electronic erase capability. The last variant in the list is flash memory. It is similar to that of EEPROM. The main difference between EEPROM and flash memory is that in the case of EEPROM, the entire memory is erased before being written, whereas in the case of flash memory, it may be erased, written, and read in blocks or bytes.

Some of the applications are as follows:

- Storing fonts for a printer
- Storing sound data in musical instruments
- Video game consoles
- Implantable medical devices
- High-definition multimedia interfaces (HDMI)
- BIOS chip in the computer
- Program storage chips in modems, video cards, and many electronic gadgets, controllers for disks, network cards, etc.

Secondary storage is a memory that is stored external to the computer. It is mainly used for the permanent and long-term storage of programs and data. Secondary memories are classified into two types. Disk drives and tape drives. Examples include floppy disks, hard disks, SD cards, pen/flash drives, SSD, and others; optical memories include CDs, DVDs, and others. Hard disk stores and retrieves data using magnetic storage. Uses multiple magnetic disks coated with magnetizable material such as iron oxide. Redundant arrays of independent disks (RAID) is another type of memory that consists of multiple disks. This technology is used to increase the performance of data storage through parallel I/O access. It provides reliable data storage. And this reliability is achieved through redundancy. SSD or solid-state disks are an alternative to magnetic disks. It is a faster memory compared to magnetic memories. Most people prefer SSD not just because it is a faster memory but it is also a green device. SSD is a green device because SSD has no moving parts and hence its energy consumption is much lower compared to the magnetic disk. SD card is a tiny flash memory card. SD cards are typically used in digital cameras and other portable devices. USB flash drive is also popularly known as pen drive. It is a plug-and-play portable storage device that uses flash memory. Optical Devices include Compact disks such as CD, CD read-write, DVD, and so on.

6.3.3 Example of Memory Unit: 4×3-bit Memory

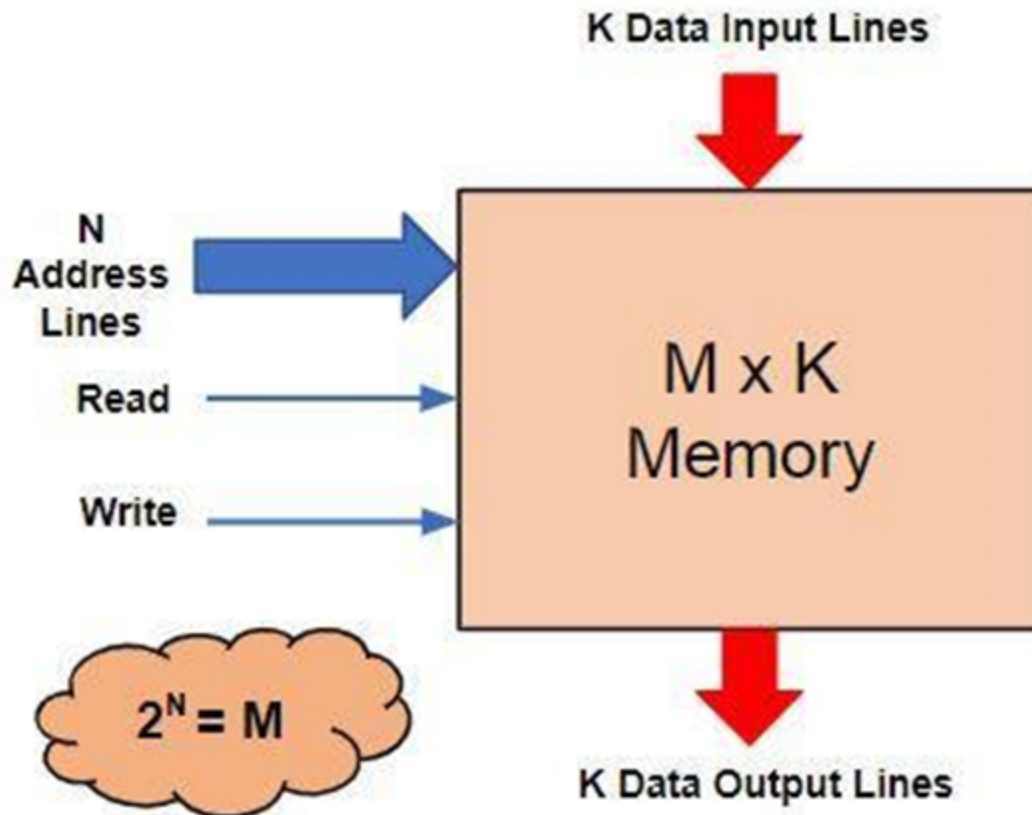


Figure 23: SRAM structure

Memory is capable of storing many 1s and 0s. Figure 23 shows the M by K read-write memory whose address space is M and addressability is K. The memory is capable of being written into and read from. Whenever the processor wants to perform reading and writing operations, it sends signals on read and write lines, respectively. To perform a read or write operation, the processor has to send an N-bit address on the address lines. The relationship between the number of address bits and the address space is given by $2^N = M$. Data to be written into is sent on the K data input lines, and data to be read from is sent on K data output lines.

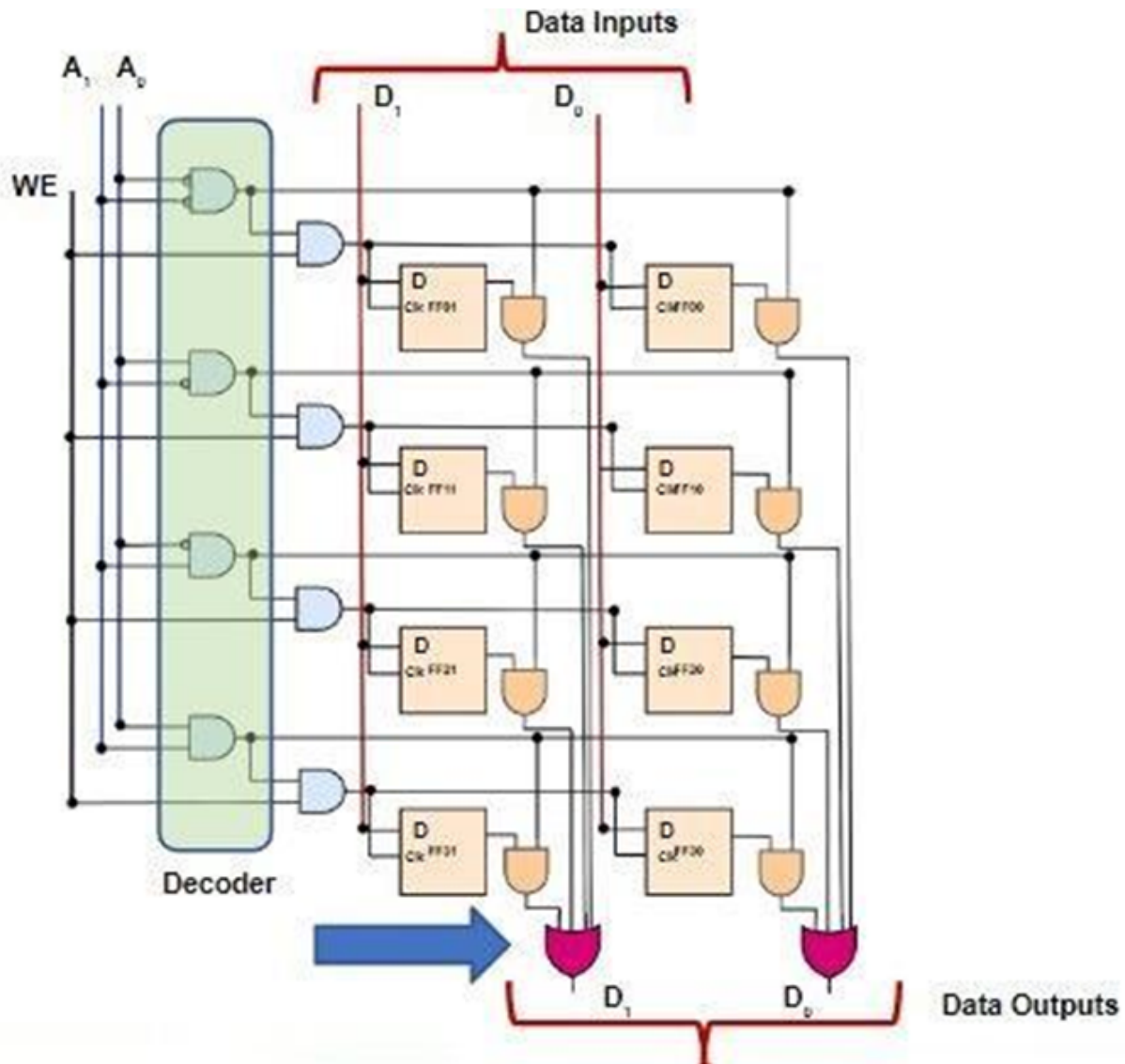


Figure 24: 4x2 memory structure

Figure 24 shows the 4x2-bit memory. There are four memory locations; in each location, 2 bits of information can be stored. Thus, a total of 8 bits of information can be stored in this memory. To store these 8 bits, for each location, two flip-flops are needed. Two address lines, A_1 and A_0 , address four memory locations. To provide the addressability of two, there are two data input lines and two data output lines. Data inputs are fed to the D input of the flip-flop. Data input gets latched into the flip-flop when the clock signal is enabled. Data inputs are written to which location is decided by the binary signal on address lines A_1 and A_0 and write enable WE line. The two address lines are connected to the 2:4 decoder circuit.

Consider an example that illustrates memory write operation. Assume that the data to be written is 1 0, and the address of the memory location is 0. So, A_1, A_0 is set to 00, and D_1, D_0 is set to 10.

To write data into the memory, write enable WE signal is set to 1. This enables the first output of the decoder, which in turn is connected to the clock inputs of the first two flip-flops, FF01 and FF00. By enabling the clock inputs, both flip-flops store data available at D_1 and D_0 inputs. In this case, it is 1, 0. In the case of a memory read operation, data is obtained from the OR gate output lines. This OR

gate is driven by the AND gates. One of the inputs for these AND gates comes from the output of the flip-flop, and the decoder output drives the other input.